Ryan Zhang

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Education

University of Waterloo | Waterloo, ON

Bachelor of Applied Science – Mechatronics Engineering | Sept. 2026 – April 2031

Technical Skills

CAD & Prototyping: SOLIDWORKS, AutoCAD, 3D Printing, Rapid Prototyping, Mechanical Assembly
Embedded Systems & Software: C/C++, Python, MATLAB, Microcontrollers, Raspberry Pi, Linux, Git, Office 365
Electronics & Hardware: PCB Design, Circuit Design, Soldering, CNC Machining, Oscilloscopes, Multimeters

Projects

Autonomous Payload Quadcopter | Jun. 2026 – Present

- Built an **F450-frame** quadcopter from brushless motors, **ESCs**, and an **11.1V 3S LiPo** pack, pairing a **Pixhawk 6C** flight controller with a **Raspberry Pi** companion computer to keep vision workloads off the flight control loop
- Engineered a **MOSFET** switching circuit (**IRF520** driver) to actuate a water-pump payload from **Raspberry Pi GPIO**, isolating logic-level signals from pump inrush current to prevent brownouts on the companion computer
- Specified a power distribution scheme around a **5V UBEC** to step battery voltage down for the Pi, separating the logic rail from high-current ESC loads and keeping onboard electronics stable through full-throttle draw
- Extending the platform with an **M10 GPS** module for **Pixhawk**-driven waypoint navigation and an **OpenCV** detection pipeline on the Pi, targeting autonomous target tracking and payload release with no operator input

Experience

Co-Founder & Lead Developer | Jul. 2026 – Present

Lunate Studios | Self-Employed

- Co-founded a studio and led development across a portfolio of **4 published Roblox games**, reaching **8M+ visits** and **10K+ peak concurrent players** while generating over **$78k in developer revenue**
- Architected persistent backend systems in **Luau** using **DataStoreService** and **OrderedDataStores**, delivering player progression, global leaderboards, inventories, and in-game economies that sustained state reliably at peak concurrency and enabled **developer product** monetization
- Instrumented custom **player analytics** and used session drop-off data to redesign onboarding, lifting tutorial completion from **24% to 70%+** and converting more first-time players into retained users
- Collaborated with my co-founders on **raycasting** and **physics-based gameplay systems**, dividing feature ownership and reviewing each other's code to ship updates on a consistent release cadence

Bicycle Technician & Youth Ride Leader | Jun. 2024 – Sept. 2026

Markham Cycles | Markham, ON

- Diagnosed and repaired mechanical faults across **braking, drivetrain, wheel,** and **tire systems** on donated bicycles, using structured inspection to isolate root causes before committing parts
- Replaced and adjusted worn components to restore bikes to safe operating condition, contributing to a team effort that refurbished **1,100+ bicycles** and distributed **800+** to over **1,000 community members**
- Led youth group rides and onboarded new volunteers on repair workflows and safety practices, sustaining shop throughput through high-volume donation periods

Robotics Team Leader | Sept. 2022 – Jun. 2026

Pickering High School | Ajax, ON

- Directed the mechanical design, fabrication, and wiring of a **VEX** competition robot, integrating an **IMU**, **distance sensor**, and a **pneumatics subsystem** through iterative prototyping that improved reliability season over season
- Programmed and tuned **autonomous** and **driver-controlled** routines using **IMU** and **distance-sensor** feedback for closed-loop correction, coordinating with teammates to resolve faults inside tight competition deadlines
- Coached new members on tool usage, wiring, and safety procedures, building a team capable of independent troubleshooting and carrying it to a **regional silver medal** finish

DECA Competitor & Team Trainer | Sept. 2023 – Jun. 2026

Pickering High School | Ajax, ON

- Placed **4th in Atlanta**, **6th in Anaheim**, and **10th in Orlando** as a **three-time ICDC Finalist**, qualifying for the international conference in three consecutive years
- Competed in DECA's **Virtual Business Challenge – Personal Finance**, applying financial modelling and business analysis to simulated case studies under timed decision cycles
- Trained newer members in case analysis, presentation delivery, and competition strategy, expanding the chapter's pool of qualifying competitors